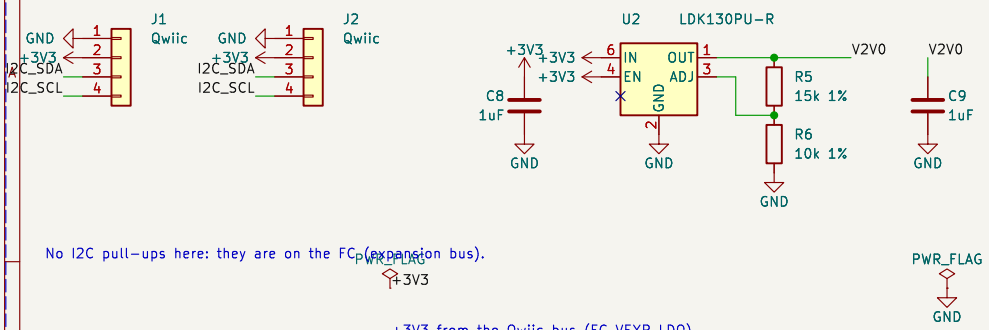
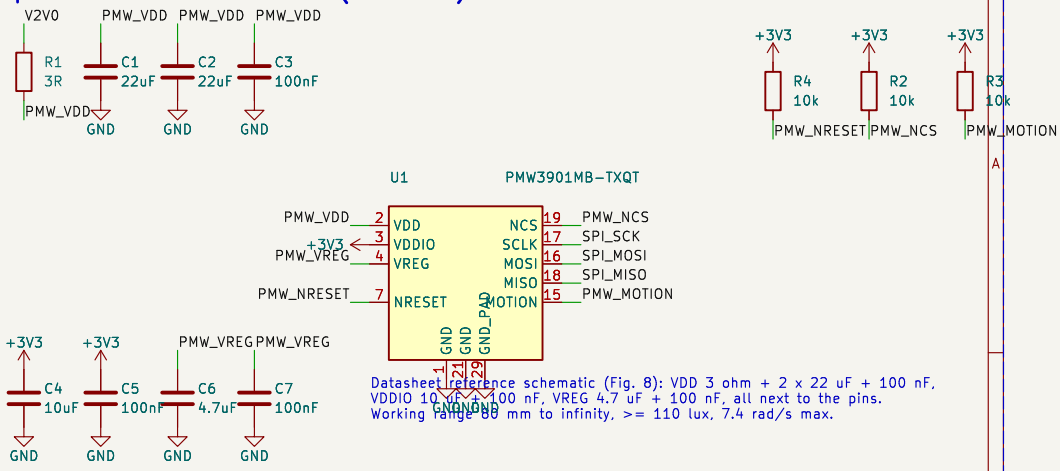
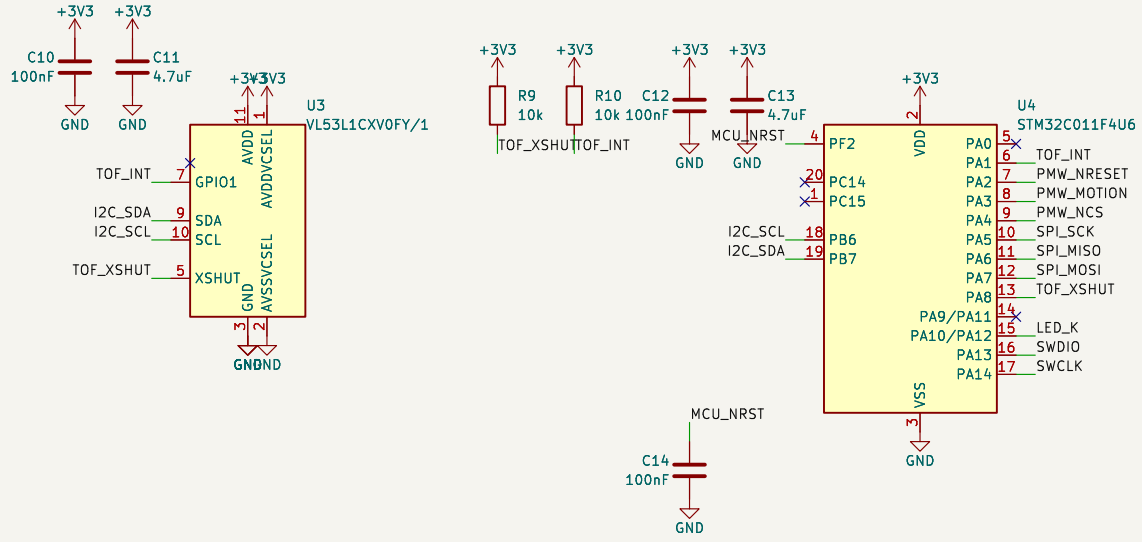
A — Qwiic in/out, 2.0 V for the flow sensor



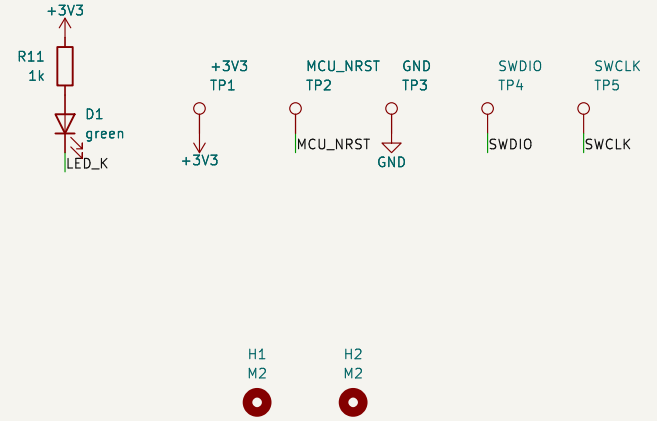
B — Optical flow PixArt PMW3901MB (SPI 2 MHz)



C — ToF ST VL53L1X (bus 0x29) and bridge MCU ST STM32C011 (bus 0x3B)



D — Status LED, SWD pads, mounting



Generated automatically: run ERC before release
PixArt PMW3901MB (flow, SPI) behind an ST STM32C011 I2C bridge; ST VL53L1X ToF on the bus
Qwiic (JST SH 4P) in/out: daisy-chain with the GPS board; I2C 0x29 (ToF) and 0x3B (flow bridge)
Mounted under the battery tray, sensor side facing the ground (handoff doc for the frame)

ACwing — Andrea Cirillo

Sheet: /
File: ACwing_Flow_A.kicad_sch

Title: ACwing Flow rev A — optical flow + ToF expansion for ACwing FC rev A

Size: A4	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 1/1